Certified Robustness for Model-Free PID Tuning by Step-Response Inspection

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Abstract

The SPIN method tunes a PID controller without a model. After each setpoint step, it counts how many turns the control error makes in three phase portraits, and it raises the gains while the counts stay below calibrated limits, so its robustness was not guaranteed. We explain why and close the gap in three steps. First, a count measures setpoint crossings, and crossings reveal the sensitivity peak M_s only on special loops: on loops that cancel the plant lags, the error never crosses the setpoint if and only if $M_s \leq 1.394$; in general they are not tied, so the counts cannot certify. Second, the step that SPIN already records contains M_s: its increments are exactly the impulse response of the sensitivity function of the digital loop, so one finite record certifies bounds on M_s without a model. Third, certified SPIN accepts a tuning only if this upper bound is at most a target M^*; so the certificate, not the counts, guarantees $M_s \leq M^*$, and the test never traps SPIN. In simulations on 19 loops, published SPIN exceeded $M^* = 1.7$ on up to 9 loops and certified SPIN on none.

Algorithm 1: one iteration of SPIN [1] and, with lines 7a–7c in bold instead of line 7, of certified SPIN (Section 4)

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 1: apply a setpoint step and record the error e
 2: if the record does not settle then
 3:     divide K_i, K_p, K_d by 2, 4, 8                    (retreat)
 4: else
 5:     count the turns N_0, N_1, N_2 of Γ_0, Γ_1, Γ_2
 6:     if N_k ≤ N̄_k for every band k then
 7:         raise all gains by one increment               (accept)
7a:     compute M̄ from the record (Theorem 2)
7b:     if M̄ ≤ M* then raise all gains                     (accept)
7c:     else lower the gains: the band move that
            Prop. 3 predicts to lower M_s (highest
            such band, at most n_max in a row), else a
            uniform back-off                               (reject)
 8:     else
 9:         k ← lowest band with N_k > N̄_k
10:         lower the gain of band k by one increment and
            raise the gains of the bands below k by one
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An increment scales a gain by $1/(1-\beta)$. Defaults: $\bar{N} = (0.5, 0.75, 1.0)$, $\delta = 0.02$, $\beta = 0.1$; gains stay within $[0.001, 10]$ times their initial values (the gain box); the result is the last accepted tuning. Certified SPIN: in line 7a, $\bar{\theta}$ solves $1 + B/\sin(\bar{\theta}/2) = M^*$ ($\bar{\theta} = \pi$ if no solution) and $h = \bar{\theta}/\lceil \bar{\theta} J/(2\eta) \rceil$, so that $hJ/2 \leq \eta$; $M^* = 1.7$, $\tau = 10^{-3}$, $\eta = 2 \cdot 10^{-3}$, $n_{\max} = 10$.

1 Introduction

Let $L = CP$ be the loop transfer function and $S = 1/(1+L)$ the sensitivity function. Its peak $M_s = \max_\omega |S|$ measures robustness: a value near 1.4 is safe, and a value near 2 indicates poor robustness [2]. Tuning rules reach a chosen M_s by using a model [2, 3]. Once a loop is tuned, M_s is rarely checked, because a check needs either a model or a test that disturbs production [5, 6].

1.1 The SPIN method

SPIN [1] tunes a PI or PID controller with the gains K_i, K_p, K_d without a model. In each iteration, it applies one setpoint step, records the control error e, and forms three phase portraits,

$$\Gamma_0 : (e, \textstyle\sum e), \quad \Gamma_1 : (\Delta e, e), \quad \Gamma_2 : (\Delta^2 e, \Delta e), \tag{1}$$

where $\sum e$ is the running sum of e and Δ is the difference between consecutive samples. The portraits $\Gamma_0, \Gamma_1, \Gamma_2$ correspond to the frequency bands of K_i, K_p and K_d, which we call bands 0, 1 and 2. The turn count N_k is the number of turns that Γ_k makes around its end point before the error stays within $\pm\delta$ (2% of the step by default). Algorithm 1 states the rule. Its limits $\bar{N}_k$ were found by calibration, and the robustness of the result was not guaranteed. An online simulator lets the reader try SPIN.[1]

1.2 Idea and contributions

The idea of this paper fits in one sentence: *the setpoint step that SPIN records in every iteration already contains M_s, so*

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[1] https://robopid-simulator.uceeb.cvut.cz

SPIN can keep its counts to shape the tuning and let the record decide whether the tuning is accepted. Three steps, each ending in a theorem, lead there.

1. *Counts cannot certify* (Section 2). A count measures setpoint crossings (Lemma 1, Corollary 1); at small δ, it measures the damping of the dominant poles (Proposition 2). Crossings reveal M_s on loops that cancel the plant lags (Theorem 1), but not in general.
2. *One step reveals M_s* (Section 3). In a digital loop, the record gives the sensitivity function exactly (Lemma 2), and a finite record gives certified bounds on M_s (Theorem 2).
3. *Certified SPIN* (Section 4) is SPIN with one more test: a tuning is accepted only if the upper bound from its record is at most a target M^*. It accepts only tunings with $M_s \le M^*$, and the test never traps it (Theorem 3).

Section 5 confirms each step in simulation.

Related work Margins have been estimated from step, relay and closed-loop data [7, 8, 9], PI tuned from one step [16], and performance optimized without a model [14, 15, 17].

2 Counts Cannot Certify

2.1 Every crossing adds half a turn

In each portrait (1), the second coordinate q is the running sum of the first coordinate p; in continuous time, $\dot{q} = p$. The turn count is the net counterclockwise angle swept around the end point, which we place at the origin, divided by 2π. We state the results in continuous time; the sampled counts of SPIN approximate them. For Γ_0, the zeros of $p = e$ are the setpoint crossings; for Γ_1, they are the extrema of e.

Lemma 1 (Half turns) *Let $\dot{q} = p$ with p continuous, let $p(t_1) = p(t_2) = 0$ with $q(t_1)q(t_2) < 0$, and let p not change sign on (t_1, t_2). Then, on $[t_1, t_2]$, the curve (p, q) turns counterclockwise around the origin by exactly half a turn. The same holds for its image under any linear map with a positive determinant, such as the whitening used in [1].*

Proof: On (t_1, t_2), p keeps one sign. Hence the curve stays in one closed half-plane bounded by the q-axis, and q moves monotonically from $q(t_1)$ to $q(t_2)$, that is, from one ray of the q-axis to the opposite ray; the curve avoids the origin, since $q(t) = 0$ only where $p \neq 0$. The angle therefore stays in a closed interval of length π whose ends are the two rays, so it changes by exactly π. If $p > 0$, q increases; if $p < 0$, q decreases; in both cases the turn is counterclockwise. A linear map with a positive determinant maps the half-plane to a half-plane and the two rays to opposite rays, and it preserves the orientation. □

Corollary 1 (Counts and crossings) *Let $p(0) \neq 0$, and let p change sign at $t_1 < \cdots < t_m$, $m \ge 0$, before the error stays within $\pm\delta$. Let $q(t_j)$ alternate in sign, and let $q(t_1)$ have the sign of p before t_1. Then the turn count of (p, q), and of its image under any linear map with a positive determinant, lies in $(\frac{m-1}{2}, \frac{m+1}{2})$. For Γ_0, these conditions hold if the areas of the lobes of e between crossings decrease strictly.*

Proof: For $m = 0$, the curve stays in one open half-plane, so its angle changes by less than π. For $m \ge 1$, by Lemma 1, each of the $m-1$ intervals between sign changes adds π. Before t_1 and after t_m, the curve stays in one open half-plane; by the sign conditions, it ends (before t_1) or starts (after t_m) on the ray from which it turns counterclockwise, so each end adds an angle in $(0, \pi)$. For Γ_0, $q(t_j) = -\int_{t_j}^{\infty} e\,dt$ is an alternating sum of lobe areas; if the areas decrease, the sum has the sign of its first term. □

2.2 When crossings reveal M_s

Lambda, IMC and SIMC tunings cancel the plant lags [3], e.g., a PI controller $K_c(1 + 1/(T_i s))$ with $T_i = T$ on the plant $Ke^{-sL_d}/(Ts+1)$, or a series PID controller with $T_i = T_1$ and $T_d = T_2$ on a plant with two lags. The loop is then

$$L(s) = \frac{k\,e^{-sL_d}}{sL_d}, \qquad k = \frac{KK_cL_d}{T_i}. \tag{2}$$

Proposition 1 (Exact robustness) *The loop* (2) *is stable for $0 < k < \pi/2$. Its gain margin is $\pi/(2k)$, and its poles are $W_j(-k)/L_d$, where W_j are the branches of the Lambert W function [10]. Its peak M_s is a strictly increasing function of k, given by*

$$k = x\sin x - x^2\cos x, \qquad \frac{1}{M_s^2} = 1 - \frac{2k\sin x}{x} + \frac{k^2}{x^2}, \tag{3}$$

where $x = \omega_p L_d \in (0, \pi/2)$ is the normalized peak frequency.

Proof: The phase of L is $-\pi/2 - x$. It first reaches $-\pi$ at $x = \pi/2$, where $|L| = 2k/\pi < 1$; since $|L| = k/x$ decreases, $|L| < 1$ at every later crossing of $-\pi$, so the loop is stable (Nyquist). The poles solve $sL_de^{sL_d} = -k$. Let $h(x) = |1+L|^2 = 1 - 2k\sin x/x + k^2/x^2$. Then $h'(x) = 2k(\varphi(x) - k)/x^3$, where $\varphi(x) = x\sin x - x^2\cos x$ increases on $(0, \pi)$ and $\varphi(\pi/2) = \pi/2 > k$. Hence, on $(0, \pi)$, h has a single minimum $x^* \in (0, \pi/2)$. On $[\pi, 2\pi]$, $h \ge 1$; for $x \ge 2\pi$, $h \ge (1 - k/x)^2 > (1 - 2k/\pi)^2 = h(\pi/2)$. So x^* is the global minimum, which gives (3). By the envelope theorem, $\frac{d}{dk}h(x^*) = \frac{2}{x^{*2}}(k - x^*\sin x^*) = -2\cos x^* < 0$. □

SIMC with $\tau_c = L_d$ and $T \le 8L_d$ cancels the lag and gives $k = \frac{1}{2}$. Hence the gain margin is π and, by (3), $M_s = 1.59$, which are exactly the values in [4, Table 2].

After a unit setpoint step, $e = 1$ on $[0, L_d]$ and $\dot{e}(t) = -(k/L_d)\,e(t - L_d)$. Let $c_1 < c_2 < \ldots$ be the setpoint crossings, let $c_0 = 0$, let lobe i be the interval $[c_i, c_{i+1}]$, and let $E_0 = 1$.

Theorem 1 (Counts on lag-cancelled loops) *For the loop* (2) *with $0 < k < \pi/2$, the following hold.*

(a) The error never crosses the setpoint if and only if $k \le 1/\mathrm{e}$, that is, if and only if $M_s \le M_s(1/\mathrm{e}) = 1.394$. For $k > 1/\mathrm{e}$, it crosses infinitely often.
(b) Lobe i contains exactly one extremum E_i, at $c_i + L_d$.
(c) $|E_{i+1}| \le r(k)\,|E_i|$, where $r(k) = k^2/2$ for $k \le 1$ and $r(k) = k - \frac{1}{2}$ for $k \ge 1$.
(d) For $k < 3/2$, the lobe areas decrease strictly, so Corollary 1 holds for Γ_0: N_0 is within half a turn of half the number of crossings.

Proof: (a) The record is the fundamental solution of the delay equation. It is positive if and only if $s + (k/L_d)e^{-sL_d} = 0$ has a real root, that is, if and only if $k \le 1/\mathrm{e}$; for $k > 1/\mathrm{e}$, all solutions oscillate [13]. The value of M_s follows from Proposition 1. (b) The sign of $\dot e(t)$ is that of $-e(t-L_d)$. By induction over the lobes, e moves away from zero on $(c_i, c_i + L_d)$, where $e(t-L_d)$ lies in lobe $i-1$; hence lobe i is longer than L_d. On $(c_i + L_d, c_{i+1} + L_d)$, e moves toward zero. For $i = 0$, the extremum is the plateau $e = 1$ on $[0, L_d]$. (c) On lobe i, $|e| \le |E_i|$. After the extremum, $|\dot e| \le (k/L_d)|E_i|$, so e needs at least L_d/k to reach zero, and $|e(t)| \le |E_i| \min\{1, k(c_{i+1} - t)/L_d\}$ on the last L_d of the lobe (for $t \ge c_i + L_d$ by integrating the bound on $|\dot e|$; for earlier t, $k(c_{i+1} - t)/L_d > 1$). Since $e(c_{i+1}) = 0$, $E_{i+1} = -\frac{k}{L_d}\int_{c_{i+1}-L_d}^{c_{i+1}} e\,dt$, and integrating the bound gives $r(k)|E_i|$. (d) Integrating $\dot e$ from $c_i + L_d$ to $c_{i+1} + L_d$ gives the area $\frac{L_d}{k}(|E_i| + |E_{i+1}|)$ of lobe i, because the extrema alternate in sign. By (c), $|E_{i+2}| < |E_i|$ if $r(k) < 1$, that is, if $k < 3/2$, where $M_s = 26.2$. □

How many of the crossings a count sees depends on δ; the limit $\delta \to 0$ removes this dependence.

Proposition 2 (Counts at small δ) *For the loop (2), let $N(\delta)$ be any of the counts N_0, N_1, N_2, in any linear coordinates with a positive determinant, which may depend on δ. If $k \le 1/\mathrm{e}$, $N(\delta)$ stays bounded as $\delta \to 0$. If $1/\mathrm{e} < k < \pi/2$, let $\sigma + j\omega = W_0(-k)/L_d$, $\omega > 0$. Then*

$$N(\delta) = \frac{\omega}{2\pi|\sigma|}\ln\frac{1}{\delta} + O(1) = \frac{\sqrt{1-\zeta^2}}{2\pi\zeta}\ln\frac{1}{\delta} + O(1), \quad (4)$$

where $\zeta = |\sigma|/|\sigma + j\omega|$ is the damping ratio of the dominant poles.

Proof: The roots of $sL_de^{sL_d} = -k$ are $W_i(-k)/L_d$; W_0 gives the rightmost, real for $k \le 1/\mathrm{e}$ and a simple pair $s, \bar s$ otherwise [11]. The residue of $\mathscr{L}\{e\} = L_d/(sL_d + ke^{-sL_d})$ at s is $1/(1 + sL_d) \ne 0$, so $e = \mathrm{Re}(ce^{st}) + O(e^{\sigma' t})$, $c \ne 0$, $\sigma' < \sigma$ [12]. In each portrait, q is the running sum of p and tends to 0, so $(p, q) \approx (\mathrm{Re}(sw), \mathrm{Re}\,w)$, $w = cs^m e^{st}$, $m \in \{-1, 0, 1\}$: the image of w, which turns at the rate ω, under a map with determinant $\omega > 0$. A map with a positive determinant changes a swept angle by less than π, so the angle at t is $\omega t + O(1)$. The count stops when $|e|$ last exceeds δ, at $t_\delta = \ln(1/\delta)/|\sigma| + O(1)$, since $|e| \le Ce^{\sigma t}$ and $|e| \ge \frac{|c|}{2}e^{\sigma t}$ once per half period; trimming moves Γ_0 by $O(\delta)$, which affects only the last $O(1)$ of the count. For $k \le 1/\mathrm{e}$, the leading term has a fixed direction, so the angle converges. □

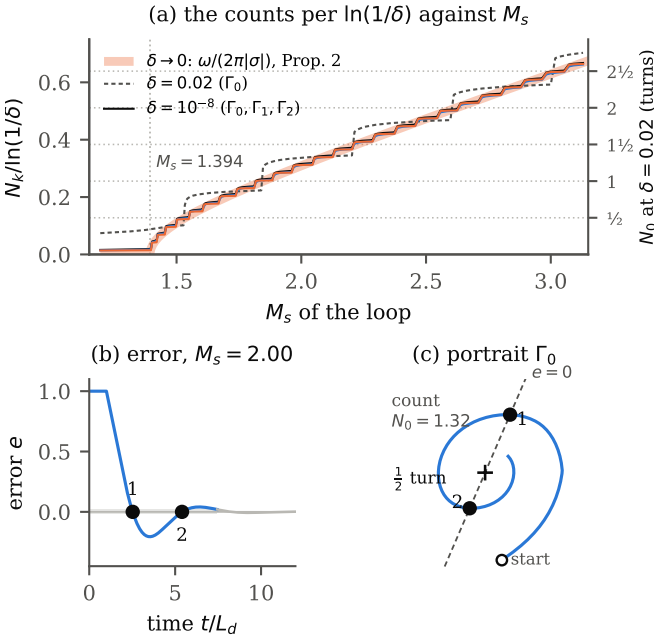


Figure 1: The lag-cancelled loop. (a) Whitened counts divided by $\ln(1/\delta)$ versus M_s: at $\delta = 0.02$ a staircase whose steps avoid the half-integer limits (dotted, right axis in turns); at $\delta = 10^{-8}$, all three counts lie on the limit (4) (Proposition 2). (b) A record with $M_s = 2.00$ and two crossings ($\delta = 0.02$). (c) Its portrait Γ_0 turns by exactly half a turn between the crossings (Lemma 1); the radius is compressed but the angle is not.

So a limit $\bar N$ at a fixed δ limits the damping of the dominant poles; the rate (4) is zero up to $M_s = 1.394$ and grows with M_s (Fig. 1(a)).

Robust limits at $\delta = 0.02$ Each crossing above δ adds half a turn (Lemma 1), so the counts form a staircase. On the loop (2), with n crossings above δ, all three computed counts lie strictly between $n/2$ and $(n+1)/2$ (Fig. 1(a), right axis). A half-integer limit $\bar N = (n+1)/2$ thus means exactly "at most n crossings above δ", that is, $M_s \le 1.534$, 1.842, 2.210 for $n = 0, 1, 2$, where the $(n+1)$th extremum equals δ. Any other limit lies on a step, and its meaning depends on the portrait: $\bar N = 0.75$ trips at $M_s = 1.54$, 1.57 or 1.59. On general loops the staircase breaks: for the plants of Section 5 with random gains, a count with m sign changes of p left $(m/2, (m+1)/2)$ in 24, 9 and 68% of 941 records $(\Gamma_0, \Gamma_1, \Gamma_2)$.

2.3 When they do not

Elsewhere, crossings and M_s are not tied. A first-order plant with $T = 0.3L_d$ under PI with $T_i = 8T$ and a gain margin of 1.624 has $M_s = 2.62$, yet its error never crosses the setpoint by more than 2%: the oscillation rides on a slow decay (Fig. 2). Conversely, a controller zero can cause crossings at a low M_s.

So the counts can shape a tuning, but they do not certify it; a certificate must, and the same step contains one.

3 One Step Reveals M_s

The counts cannot certify M_s, but the record they are computed from can. A digital controller runs with the sampling period T_s. Seen at the samples, the loop it closes is linear and time invariant. From here on, $L(z)$, $P(z)$, $S = 1/(1+L)$ and $M_s = \max_{\theta\in[0,\pi]} |S(e^{j\theta})|$, with $\theta = \omega T_s$, refer to this digital loop, the loop that actually runs.

Lemma 2 (Exact identity) *Let the loop be stable, have one degree of freedom, and have a strictly proper L (the measurement precedes the control action). Let e_n be the error after a unit setpoint step at $n = 0$, applied from rest and without disturbances, so that $e_{-1} = 0$ and $e_0 = 1$, and let $a_n = \Delta e_n = e_n - e_{n-1}$. Then*

$$S(e^{j\theta}) = \sum_{n\geq 0} a_n\, e^{-jn\theta}. \tag{5}$$

Proof: The z-transform of e is $S(z)/(1-z^{-1})$, so $S = (1-z^{-1})\mathscr{Z}\{e\}$. In a stable loop, a_n decays exponentially, so the series converges on the unit circle. □

Assumption 1 (Long enough record) *The loop is stable, and the record $e_0, \ldots, e_N$ satisfies $\sum_{n>N} |a_n| \leq \tau$ for a known τ.*

In a stable loop, $|a_n| \leq c\rho^n$ with $\rho < 1$, so τ falls geometrically with the record length.

Theorem 2 (Certified bounds from one record) *Let the conditions of Lemma 2 and Assumption 1 hold, and let $\hat{S}(\theta) = \sum_{n=0}^{N} a_n e^{-jn\theta}$. Let G be a uniform grid on $[0, \bar{\theta}]$, $\bar{\theta} \leq \pi$, that has spacing h and contains both ends. For any n_c, let*

$$J = \textstyle\sum_{n=0}^{N} |n - n_c|\,|a_n|, \;\; B = \sum_{n=1}^{N-1} |a_{n+1} - a_n| + |a_N| + 2\tau.$$

Then $\underline{M} \leq M_s \leq \overline{M}$, where

$$\begin{aligned} \underline{M} &= \max_{\theta\in G} |\hat{S}(\theta)| - \tau, \\ \overline{M} &= \max\Big\{ \max_{\theta\in G} |\hat{S}(\theta)| + \tfrac{hJ}{2} + \tau,\; 1 + \tfrac{B}{\sin(\bar{\theta}/2)} \Big\}, \end{aligned} \tag{6}$$

and the second term is omitted if $\bar{\theta} = \pi$. If τ equals the true remaining variation, $N \to \infty$, $h \to 0$ and $\bar{\theta} = \pi$, then both bounds converge to M_s.

Proof: By (5), $|S - \hat{S}| \leq \sum_{n>N} |a_n| \leq \tau$ at every θ, which gives $\underline{M}$. On $[0, \bar{\theta}]$, $|\hat{S}|$ equals $|e^{jn_c\theta}\hat{S}|$, whose derivative is bounded by J. Every θ lies within $h/2$ of a grid point, which gives the first term of $\overline{M}$. On $[\bar{\theta}, \pi]$, let $z = e^{-j\theta}$ and $Z_n = \sum_{m=1}^{n} z^m$, so that $|Z_n| \leq 2/|1-z| = 1/\sin(\theta/2)$. Summation by parts gives $\sum_{n\geq 1} a_n z^n = \sum_{n\geq 1} (a_n - a_{n+1}) Z_n$. Since $\sum_{n\geq N} |a_n - a_{n+1}| \leq |a_N| + 2\tau$, we obtain $|S - a_0| \leq B/\sin(\theta/2)$ with $a_0 = 1$. Convergence follows because, for fixed n_c, J stays bounded as $N \to \infty$, so $hJ \to 0$, and the remaining variation tends to zero. □

All of (6) but τ comes from the record, the grid values from one chirp z-transform; B bounds $[\bar{\theta}, \pi]$ at once.

4 Certified SPIN

Certified SPIN is SPIN with one more test: a tuning that the counts accept is accepted only if the upper bound $\overline{M}$ of Theorem 2, computed from the same record, is at most a target M^*. In Algorithm 1, it replaces line 7 by the bold lines 7a–7c; everything else is unchanged. The counts shape the tuning; the certificate decides whether it is accepted. When the bound rejects a tuning, certified SPIN lowers the gains in one of two ways: (1) a band move, as in line 10, or (2) a uniform back--off, the reverse of an acceptance.

Theorem 3 (Certified SPIN) *Let the plant be stable, let $K_i P(1) > 0$, and let every record tested in line 7a meet the conditions of Lemma 2 and Assumption 1. For (iii) and (iv), also let no gain be clipped by the gain box, and let $\beta < \frac{1}{2}$.*

(i) Every tuning that passes the test of line 7b, and hence the result of certified SPIN, has $M_s \leq M^*$.

(ii) For a controller C, let $\mu = \min\{0, \inf_{0<\theta\leq\pi} \mathrm{Re}\, L(e^{j\theta})\}$. For $0 < \lambda < 1/|\mu|$, the loop with the controller λC is stable, and

$$M_s(\lambda C) \leq \frac{1}{1 - \lambda|\mu|}. \tag{7}$$

(iii) If $M^* > 1 + 2\tau + \eta$, every run of consecutive rejections by the test of line 7b ends, either in an acceptance or in a move of Algorithm 1 (line 3 or line 10). The run contains at most $n_{\max}$ band moves, and each stretch of uniform back-offs in it has at most $\lceil \ln \frac{|\mu| M'}{M'-1} / \ln \frac{1}{1-\beta} \rceil$ of them, where $M' = M^* - 2\tau - \eta$ and μ belongs to the controller at the start of the stretch.

(iv) Every move except an acceptance lowers $\Phi = \ln K_i + 2\ln K_p + 4\ln K_d$ (without the last term for PI) by at least $\ln \frac{1}{1-\beta}$. Hence the gains never return to earlier values without an acceptance in between.

Proof: (i) This follows from Theorem 2. (ii) L is bounded on the unit circle except near $z = 1$. There, the integral term has a bounded real part, because P is analytic at $z = 1$ and $\mathrm{Re}\, 1/(e^{j\theta} - 1) = -\frac{1}{2}$; so μ is finite. Then $|1 + \lambda L| \geq 1 + \lambda\, \mathrm{Re}\, L \geq 1 - \lambda|\mu| > 0$, which gives (7). The Nyquist curve of λL lies in $\mathrm{Re} \geq -\lambda|\mu| > -1$, also on a small enough indentation outside $z = 1$, where $K_i P(1) > 0$ keeps $\mathrm{Re}\, L > \mu$. It therefore cannot encircle -1, and since L has no poles outside the unit disc, the loop is stable. (iii) Line 7c allows at most $n_{\max}$ band moves. In a stretch of uniform back-offs from a controller C, $\lambda = (1-\beta)^j$ after j back-offs, and (7) gives $M_s \leq M'$ for the stated j. In (6), the grid term is then

Table 1: M_s of the digital loop for the most aggressive tuning that published and certified SPIN accept, from the cautious and the bold start, and the change of the load-step IAE (cautious start; –: tuning unchanged).

		published		certified		
		caut.	bold	caut.	bold	ΔIAE
P_1	PI	1.46	1.46	1.46	1.46	–
	PID	1.66	**1.84**	1.66	1.61	–
P_2	PI	1.64	1.67	1.64	1.67	–
	PID	1.59	**1.78**	1.59	1.66	–
P_3	PI	1.67	1.63	1.67	1.63	–
	PID	**1.81**	**1.74**	1.69	1.60	+23%
P_4	PI	1.55	1.60	1.55	1.60	–
	PID	1.61	1.55	1.61	1.55	–
delay-dom.	PI	1.68	1.63	1.68	1.63	–
lag-dom.	PI	**2.01**	1.69	1.67	1.69	+52%
2 lags	PI	1.44	1.47	1.44	1.47	–
fast	PID	**1.89**	**1.98**	1.66	1.69	0%

at most $M_s+\tau+\eta+\tau\le M^*$, and the second term is at most M^* by the choice of $\bar\theta$; so the test passes unless the counts reject first. (iv) With $w=(1,2,4)$, a band move changes Φ by $\ln\frac{1}{1-\beta}(\sum_{j<k}w_j-w_k)\le-\ln\frac{1}{1-\beta}$. A uniform back-off changes it by $-7\ln\frac{1}{1-\beta}$, and line 3 by $-17\ln 2<-\ln\frac{1}{1-\beta}$ for $\beta<\frac12$. Returning to the same gains requires $\sum\Delta\Phi=0$, hence an acceptance, which raises Φ. □

So the test never traps SPIN. A back-off is safe by (ii) but slows load rejection; a band move is often cheaper, and the record predicts it.

Proposition 3 (Prediction from one step) *For any controller C' on the same plant, $S'=S/(S+(1-S)C'/C)$ exactly. If the record gives $\hat S=S+\varepsilon$, the predicted $\hat S'$ has the error $\hat S'-S'=\varepsilon\,(C'/C)/(D\hat D)$, where $D=S+(1-S)C'/C$ and $\hat D$ is D with $\hat S$, wherever $D,\hat D\neq 0$. The formula predicts the frequency response of the new loop, not its stability.*

Proof: $L'=LC'/C$ with $L=(1-S)/S$; subtracting the two expressions gives the error. □

5 Simulations

Setup We compare published and certified SPIN. Both run 200 iterations with a derivative filter ratio of 10 and the gain box of [1]. Since the box ties the result to the start, every loop runs from SIMC on a half-rule model with τ_c three times its dead time (cautious start) and from the same tuning with all gains doubled (bold start). The 19 test loops are (1) the four plants of [1], $P_1=e^{-s}/((10s+1)(s+1)^3)$, $P_2=1.25e^{-8s}/(5s+1)^4$, $P_3=e^{-10s}/((2s+1)(s+1)^3)$ and $P_4=e^{-4s}/(8s+1)^6$, with PI and with PID; (2) four plants that place the oscillation in a chosen band: the delay-dominant $e^{-5s}/(s+1)$ and the lag-dominant $e^{-s}/(10s+1)$ and $e^{-0.5s}/((10s+1)(s+1))$ with PI, and the fast $e^{-0.1s}/((s+1)(0.5s+1))$ with PID; (3) the family $e^{-L_ds}/((s+1)(0.5s+1)(0.25s+1))$ with PID and $L_d\in\{0.25,0.5,1,\dots,16\}$ (dead-time share 0.125 to 0.90). P, I and D act on the error. Let T_a be the dead time plus the lag time constants of the plant and T_f the derivative filter time constant; the controller runs at $T_s=\min(T_a/100,T_f/20)$ for PID and $T_a/1000$ for PI, with exactly discretized lags and the delay rounded to whole samples. Each record covers $15\,T_a$. All values of M_s are those of this digital loop, computed from its exact L. Because SPIN ends in a cycle between a few tunings, we report the most aggressive tuning accepted in the last 60 iterations.

Counts (Section 2) On the lag-cancelled loop, for 126 values of $k\in(1/\mathrm{e},3/2)$, Theorem 1(c), (d) and Lemma 1 held (half turns within 10^{-6}), and (3) matched a direct search to six digits. For $\delta\in[10^{-12},10^{-3}]$, the counts grew with the slope (4) within 3.3% (remainder <0.45 turn), and for $k\le 1/\mathrm{e}$ varied by <0.03.

Bounds (Section 3) For all 146 tunings accepted below, a four times longer record showed a remaining variation $\le 2.8\cdot 10^{-5}$, so Assumption 1 held with $\tau=10^{-3}$; (5) held up to this variation, and every M_s lay within its bounds, with $\overline{M}-M_s\le\eta+\tau+2.8\cdot10^{-5}$. Short records give wide bounds that still contain M_s (Fig. 2(b)).

Certified SPIN (Section 4; Table 1, Fig. 3)

- *Safety.* Published SPIN ended above $M^*=1.7$ on 9 of the 19 loops from the cautious start (up to 2.01) and on 8 from the bold start (up to 2.09); certified SPIN ended at or below M^* on all of them (at most 1.69), as Theorem 3(i) guarantees.
- *Without the gain box.* Published SPIN exceeded M^* on 12 of the 18 loops that kept accepting (up to 2.05); certified SPIN on none (up to 1.68).
- *Back-off, prediction.* (7) held along each back-off; predicted and true M_s agreed within 10^{-7}.
- *No trapping.* Runs of rejections had at most 4 band moves; every loop kept accepting, except $L_d=16$ without the box, where the counts lower K_d every round under either rule, as Theorem 3(iv) allows.
- *Cost.* Load-step IAE: +7.8% on average (+6.4% bold).

6 Conclusion

The setpoint step that SPIN records in every iteration already contains M_s. (1) Counts cannot certify: they measure crossings and, at small δ, damping; at $\delta=0.02$ the robust limits are the half-integers, which bound M_s only on special loops (Theorem 1, Proposition 2). (2) One step reveals M_s: in a digital loop, one finite record certifies M_s without a model (Theorem 2); so does every logged setpoint

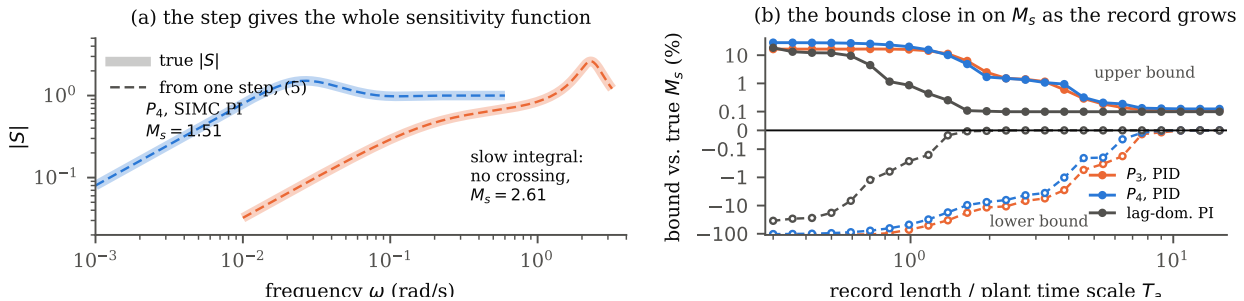


Figure 2: (a) $|S|$ from one step (dashed) lies on the true $|S|$ (thick), also where the counts miss the peak (Lemma 2). (b) The bounds (6) versus record length (Theorem 2).

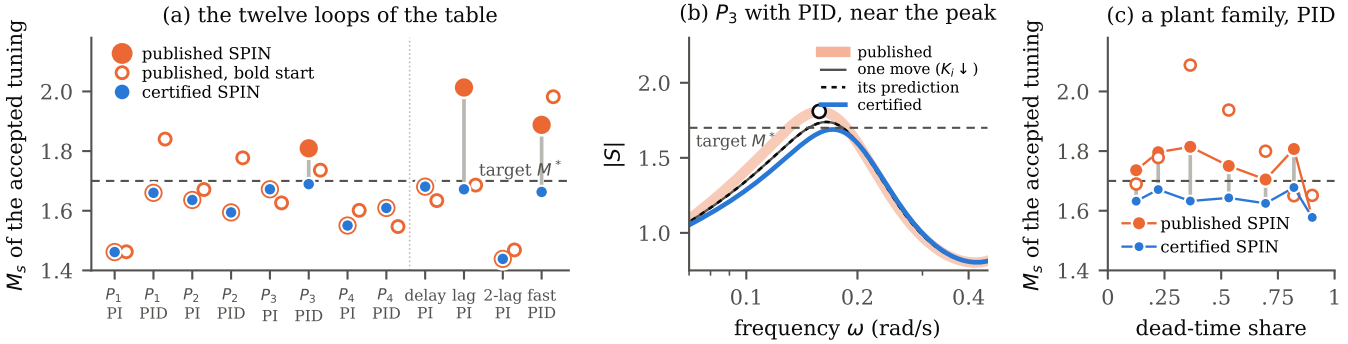


Figure 3: Published vs. certified SPIN, $M^* = 1.7$. (a) Loops of Table 1. (b) P_3, PID: published ($\overline{M} = 1.81$, ○), one move lowering K_i (1.739, Prop. 3), certified (1.689). (c) Plant family.

change made from rest. (3) Certified SPIN, with this certificate as its acceptance test, accepts only tunings with $M_s \le M^*$ and is never trapped (Theorem 3). Not the counts certify a tuning, but the certificate. The theorems assume a linear loop and a long enough record; noise and real loops are left to a follow-up. Scripts: `https://github.com/ottapav/robopid-simulator` (`docs/ECC27_certified_SPIN`).

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